

Mobile communications for an operational field force

J Garrett

This paper reviews BT's operational need for mobile communications and some of the factors that are being considered in moving to an integrated solution that will meet the needs of most of their field force for some time to come.

1. Introduction

Mobile radio is used by utility and service organisations to provide control and communications for their field engineers and other mobile workers. Historically BT has not made much use of mobile radio except for the external plant maintenance part of the field force and now has a number of private mobile radio (PMR) systems on VHF low band at over 200 sites throughout the UK, providing coverage similar to that found in some cellular telephone networks. Efforts to upgrade the existing systems and achieve better use of them continue in parallel with a new programme to provide mobile communications to all the field force. The most visible step has been the large-scale provision of cell-phones as an interim measure, though this should not necessarily be seen as the long-term solution.

2. The users

The users can be divided into the following three main groups.

- Customer premises equipment installation and repair, serving both personal and business customers require data communications for Work Manager [1] terminals and other job despatch arrangements, together with voice for general enquiries and assistance.
- Underground and other external plant maintenance teams have usually had radio systems to provide voice only links with their control points. Many controls have closed or moved and a more flexible radio system would be desirable.

- Field managers, now mainly mobile rather than desk-bound, have a need for access to several of the main-frame computer systems that provide management information, e-mail and other employee communications.

There are other potential users in logistics and core switching and transmission operations.

The requirements range from voice, short data messages — well suited to GSM short message service (SMS) or PMR short data message (SDM) — longer file transmissions, through image/facsimile to interactive terminal sessions.

Nearly all user groups state voice as an essential feature. For dealing with exceptions and, of course, for talking to customers, this is the case, but data is the preferred option for routine traffic, be it job control or employee communications, as data transactions generally allow better use of the radio channels. Data also fits in well with current systems for job despatch and progress monitoring by computer.

Typical job despatch messages, as used with Work Manager [1], are about 3 kbyte long. There are pressures to increase this to provide more information; however, when being sent by radio it is necessary to compress the data sent and so save transmission time and costs. Of course, Work Manager is designed to send, in one shot, everything that

might be needed as there is only one chance to do so, whereas, with a good mobile data system, users can make enquiries as and when they need extra information, so it should be possible to send substantially less data for run-of-the-mill jobs.

For jobs that require, say, cable diagrams, the radio system must provide for rapid transfer of even more data, a typical diagram requiring maybe 35 kbyte of data if sent as a compressed file. Alternatively, facsimile, which applies its own compression techniques, may be used. Cellular supports 9.6 kbit/s fax transmission, while slower speeds are used over duplex PMR, sending a typical A4 page in about one minute.

3. Solutions being trialled

Before any national roll-out of a solution it is normal to carry out small trials, perhaps of several techniques, then a pilot to check that the whole operation, including installation, training and operational matters, is workable. The aim is to provide reliable systems that are user-friendly and well integrated with the office systems and processes that interface to them. In this respect, voice-only cellphones provide very little of a solution for Work Manager-based people. Early trials of cellular data with Work Manager, BT's job despatch and control system for repair and installation work, showed that there are several advantages in having at least van-based mobile data capability, even if fully portable cellular data terminals are not yet available.

Trials have included cellular voice, both TACS and GSM, PMR with data (including revertive data paging, on which more is said later in section 4), RAM two-way paging, and GSM data.

Readers will be familiar with cellular voice service, but may not have used GSM data services (see also Fenton et al [2]) nor modern PMR or RAM mobile data so a brief description of these follows.

GSM permits two-way data up to 9.6 kbit/s, facsimile and SMS (short message service permits text messages of up to 160 characters to be sent and received and, if required, stored on suitably equipped phones). While most of these services currently require a phone plus at least one other device, integrated terminals such as the Nokia Communicator 9000 are becoming available now and will be attractive for some mobile applications.

Modern PMR systems can support voice and data and, rather like SMS, short data messages that can be sent or received by mobiles. As with SMS, these messages are automatically acknowledged, giving some comfort to the sender that the message has at least reached the destination terminal, even though the recipient may not have yet read nor acted upon it.

RAM Mobile Data operate a Mobitex [3] network of about 250 base stations providing a two-way packet data service with automatic acknowledgement between suitable terminals. For some despatch operations it can be ideal, allowing use of a hand-held integrated terminal to display messages and permitting the mobile user to send pre-programmed responses to the controller. Radio modems operating to the Mobitex standard are also available either as stand-alone devices to be used with note-book computers or as small units that may be incorporated into, say, ruggedised portable terminals. With portable terminals operating at 2 W transmitter power on UHF the service provides a useful alternative to other public systems and Ram Mobile Data has recently been selected as a main carrier of mobile data for the Home Office. One of its strengths is that the Mobitex standard is mature enough for a number of terminals and applications to have been developed.

4. The cellular versus PMR debate

There has been much pressure to use cellular, because of BT's involvement in Cellnet. Cellnet is, indeed, a potential solution for much of the traffic but there are concerns with coverage, control and cost of using cellphones. Each of these issues is discussed below.

4.1 Coverage

PMR has attractions in terms of coverage in rural areas simply because the signals at the frequency used (80 MHz) propagate with a lower loss for a given distance than does cellular at 900 MHz. In hilly terrain, where illumination of valleys may be poor at 900 MHz, low-band VHF offers advantages. However, in the urban areas, man-made noise is much more prominent at 80 MHz than at 900 MHz. Building penetration may also be better at the higher frequencies. Thus there are horses-for-courses in the choice of frequency to achieve coverage.

The need is to contact the field workers whether they are in their vans or working on a job. A typical installer may be in the van only 30% of the day — indeed, the less time in the van, the more time for productive work! Mobile communications need to get to and from the user, not just the vehicle. Recognising that neither cellular nor PMR will provide adequate in-building coverage for reliable communication with people on a job, plans are made to use a vehicle-based relay for two-way data to the user's hand-held terminal. This eases the requirements for terminal-mounted radio modems as the power required for transmission to a vehicle parked, say, 100 m away is much less than that needed to work to a base station several miles distant.

4.2 Control

Users of PMR systems frequently need to have control of their own system so that in the event of a local crisis, for example, when public mobile systems may be congested, utility or emergency services may continue operations. It is technically possible to provide priority service on cellular systems but there has, so far, not been any commercial arrangement to support this in the UK. Indeed, public mobile systems do not offer any guaranteed level of service availability.

By providing their own networks, utilities may equip the system with their chosen level of redundancy and provide other precautions against failure, e.g. high-capacity batteries or stand-by generators at isolated sites. If a site becomes cut off from the switching system it is usual for service to fall back to local 'repeater' mode operation. Cellular systems generally have little endurance in the absence of mains power and may offer no service if a site becomes isolated because of loss of the link between base stations and the rest of the network.

4.3 Costs

The most obvious advantage of using cellular is that the user does not need to make a large capital investment in the network. Cellular telephones can appear to be low priced, especially when bundled with an air-time agreement. However, the call charges are significant and, though charges have fallen recently, there are types of communication needed in the field that are not good news when one is paying by the second. For example, when jointers in adjacent man-holes 200 m apart need to have an open line for prolonged periods it is very expensive to use cellular, when PMR could meet the requirement. There is a danger that, because BT is providing cellphones to field workers they will be used where other radios would be better. Analysis of the comparative costs is helping to determine whether cellular or PMR is appropriate for any particular function.

4.4 Other features

Some of the desirable features of a radio system for field use include very rapid call set-up times, which may be essential for safety in some situations. Imagine trying to call a winch operator for an emergency stop by a cellular telephone call with a call set-up time of several seconds. This is clearly an application for PMR at present.

There are times when a group call or broadcast to all mobiles can be useful. This is, at present, not available on a cellular system.

4.5 Summary of cellular versus PMR

While many of BT's requirements for field radio communication may be met by cellular telephones, there are some that cannot and even for those that can, costs may be prohibitive. For talking directly to customers then cellular, in good coverage, is better than current PMR, but for the bulk of communications with 'control', PMR has the advantages of low cost and a reliability that is under BT's control.

5. Technology for private mobile radio

PMR has been in use for many years with a wide range of frequencies and equipment types. Standards exist which ensure that equipment will not cause interference to other services but, within these standards, there is scope for much variation in implementation of systems.

The bulk of BT's existing PMR systems are in VHF low band, where BT has six channels for nation-wide exclusive use (see Fig 1). Each channel consists of a pair of frequencies, separated by 13.5 MHz, with the higher frequency being used for base station transmitters, the lower for mobiles. Most of the equipment uses amplitude modulation (AM), though newer systems use frequency modulation (FM), which offers better performance in the presence of impulsive noise, such as vehicle ignition interference, for example.

For short-range use, up to a mile or so, BT has some VHF channels that are used with FM equipment. Examples of this are the Transceiver 5, 9A, 10A and Equipment Radio 15. These radios operate simplex and transmit and receive on the same frequency.

There are also a number of security and on-site systems operating in the UHF band, using FM equipment.

For job allocation and control applications it is advantageous to have data transmission with a display or print-out at the mobile, so that job details can be passed without the errors that might occur with voice transmission and hand-written copy. Most of BT's systems are licensed for data use, though very few are so used at present. An exception to this is the very successful Radio PSION system run by Business Systems in the major cities in Scotland. Using readily available and cheap equipment, this provides reliable two-way data over a packet radio transmission system, with printing of job details on a mobile terminal installed in the vehicles. Encryption of customer details is used to prevent unauthorised reception.

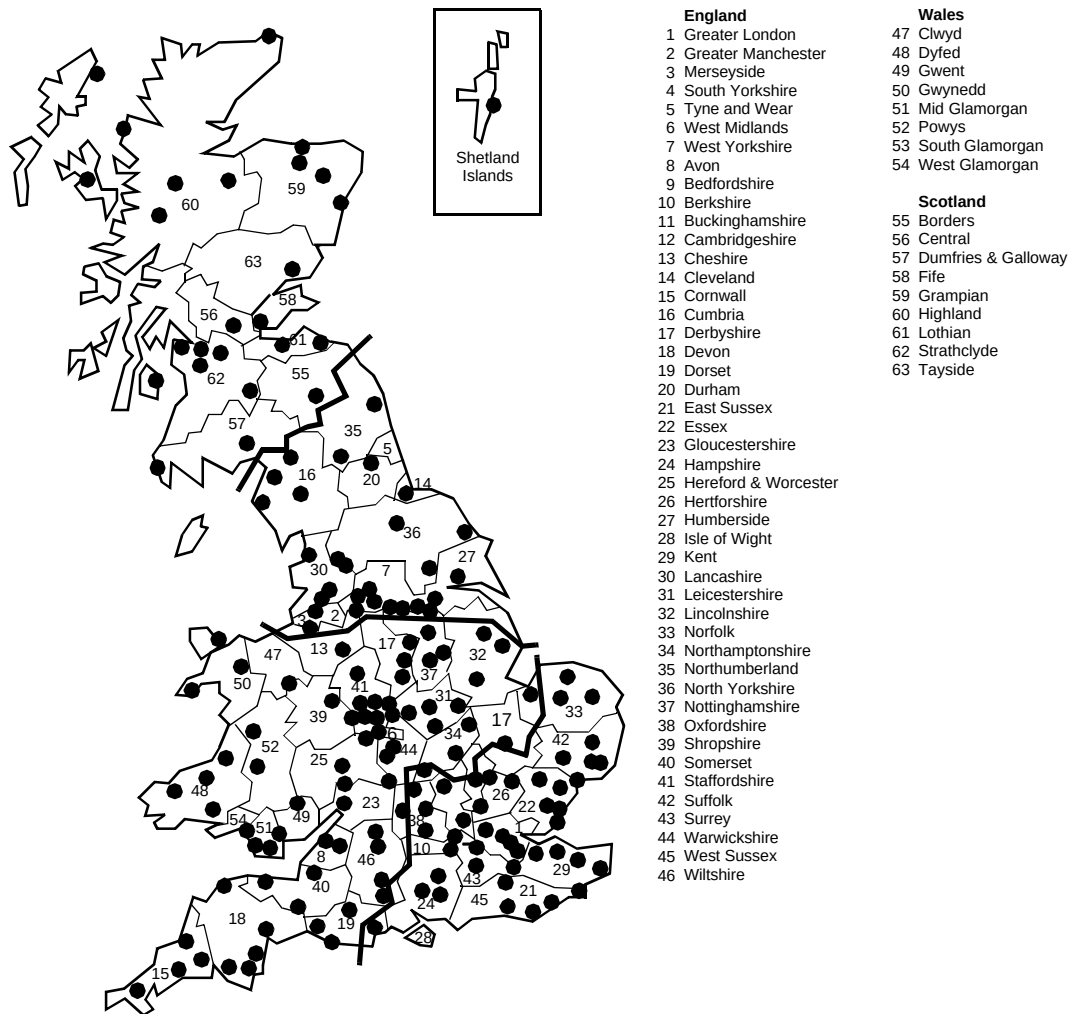


Fig 1 BT's PMR base station locations.

6. Spectrum availability and modulation methods

PMR requires that BT has its own nation-wide spectrum with sufficient channels to make a workable, integrated system that will carry the traffic in all but the dense urban areas, where cellular would be appropriate. Band III, sub-band 3 is seen as a desirable and, possibly, available frequency band, offering a good compromise between the wide-area coverage of low-band VHF on the one hand and

low-noise UHF, which, by virtue of its short wave length, is handy for portable equipment. Recently the Radiocommunications Agency announced its plans to open up new channels in this sub-band for PMR and public access mobile radio services, on 12.5 and 6.25 kHz channel spacing. Equipment for the latter is not currently available, though technology developed for 5 kHz channels (see Linear Modulation (LM) below) could be used with minimal modification (see Fig 2 for existing allocations).

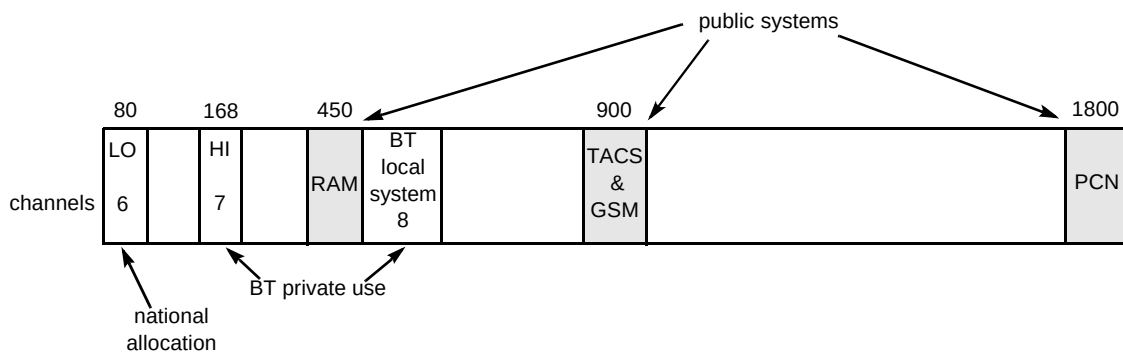


Fig 2 Existing spectrum allocation.

While it is expected that existing analogue FM equipment would have to be used initially, the arrival of TETRA (trans-European trunked radio) [4], the new digital standard for PMR, is only a year or two away. Manufacturers are already demonstrating and testing mobile equipment. Its advantages in terms of spectrum efficiency, data throughput, voice security and system features will make it the system to adopt where possible. Negotiations with the Radiocommunications Agency are taking BT's need for transition to TETRA into account.

TETRA offers twice as many channels/Hz bandwidth as existing 12.5 kHz FM with the added benefits of duplex operation by time division and the intrinsic security against casual eavesdropping of digital modulation. For users, such as the emergency services, further encryption is available should it be required. Data services, both circuit-switched and packet, are provided for, with data rates up to 7.2 kbit/s or, by using concatenated time-slots, 28.8 kbit/s (see Fig 3 for the structure of TETRA TDMA frame)

An alternative to conventional FM modulation is available in the form of the narrow-band system called Linear Modulation by its developers, Securicor [5]. This is a single-sideband (SSB) modulation scheme with added pilot tone in band and uses fast digital signal processing techniques to manipulate the audio and to linearise the transmitter power amplifier. The pilot tone serves as a frequency reference, permitting frequency control to compensate for drift and Doppler shift, which have a severe effect on SSB. LM has been designed for deployment in channels spaced at 5 kHz, though it now seems as if the UK 'narrowband' channels will be on 6.25 kHz spacing. Data

transmission at rates up to 14.4 kbit/s has been demonstrated, though usable throughput is dependent upon signal quality and may typically be half this rate. Even so, this is considerably faster than is achieved on most FM systems, even in 25 kHz channels. At present the main manufacturer of LM equipment is Securicor, though some US manufacturers have been licensed. It is seen as a commercially difficult choice, despite its possible technical benefits and the potential spectrum availability. Figure 4 compares the bandwidth of different mobile services.

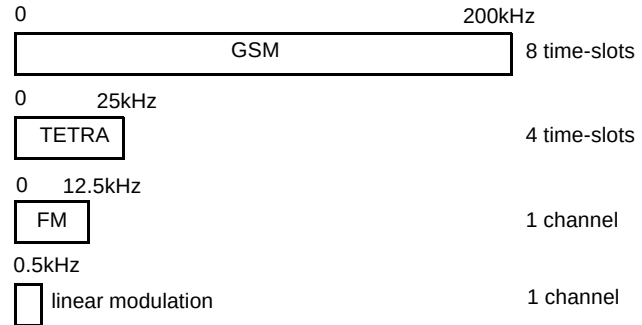


Fig 4 Comparison of mobile service bandwidths.

7. Integration with computing systems

The value of mobile data solutions is that they permit field workers to access the BT systems that support them, such as Work Manager, customer service system (CSS) and e-mail, depending upon the work area. Plain voice, while handy for many situations, is not seen as the cost-effective way to use mobile systems for normal operation.

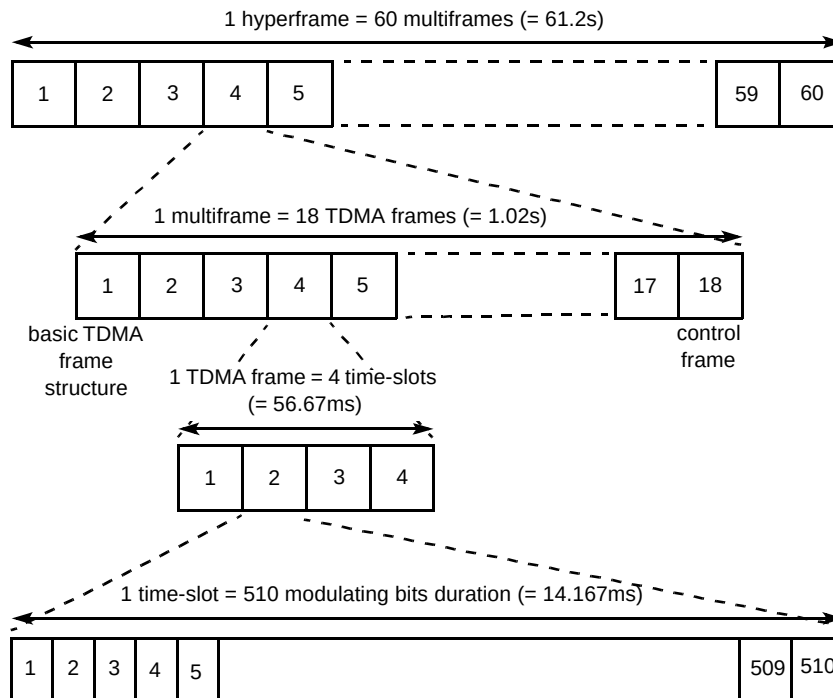


Fig 3 TETRA TDMA frame structure.

Access to computer systems from mobiles requires authentication and encryption to ensure that security of customer information is preserved and unauthorised users excluded. Gateways have been designed to provide these services, using some of the standard features of Windows NT for most main-frame applications. Work Manager has its own field access interface sub-system, which is being upgraded to deal with the mobile access requirements.

8. Feel-good factors

From being a low user of mobile communications for customer-facing field force, it is now BT's aim to provide the tools that will enable them to do their work more easily and efficiently. But there are drivers beyond the simple business efficiency needs. Mobile communications will also help safety and personal security, especially important now the amount of after-dark work is increased. On the down-side, though, mobile radios, especially portable cell-phones, are targets for theft so security needs to be good.

9. Conclusions

Mobile communications offer efficient, flexible methods to maintain contact with the field force and direct work to ensure that BT's customers obtain the high level of service that they rightly expect. This paper has outlined some of the activities directed towards improving such communication.

References

- 1 Laithwaite R: 'Work allocation challenges and solutions in a large-scale work management environment', *BT Technol J*, **13**, No 1, pp 46—54 (January 1995).
- 2 Fenton C J et al: 'Mobile data services', *BT Technol J*, No **14**, No 3, pp 92—108 (July 1996).
- 3 Larsen M: 'Mobitex — a system for mobile data communication', *Information Network and Data Communication III*, Proc IFIP TC6 International Conference (1990)
- 4 Nokia Telecommunications: 'Overview of Nokia TETRA System', internal document number: NTC NTETSYS.
- 5 'Linear Modulation', *Land Mobile*, pp 6—7 (June 1994).



John Garrett graduated from Cambridge University in 1969 with a degree in Electrical Sciences. He joined BT's Research Department working on a variety of switching systems.

From 1979 he worked on the Radiophone System 4 and later on the evaluation of cellular systems world-wide. Between 1983 and 1990 he was with Cellnet, responsible for system quality.

He is currently Senior Advisor, Mobile Systems, at BT Laboratories, on cellular quality investigative work, field communications, and PMR system evaluation.